

ERNEST ADJEI ADJETEY

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Education

California Institute of Technology

Sept 2024 – June 2028

B.S. in Computer Science & Information and Data Science

Pasadena, CA

Relevant Coursework

- Machine Learning
- Statistical Inference
- Computer Architecture
- Embedded Systems
- Data Structures & Algorithms
- Computer Vision
- Database Systems
- Operating Systems

Technical Skills

Languages: Python, Java, C/C++, SQL, JavaScript, Bash, MATLAB

Frameworks & Libraries: React, Node.js, Express.js, Flask, FastAPI, PyTorch, TensorFlow, Pandas, NumPy

Databases & Cloud: AWS Neptune, PostgreSQL, MySQL, MongoDB, Firebase

Developer Tools & Technologies: Git, Docker, Linux, Maven, JUnit, Pytest, Jira, REST APIs, Gremlin, MCP

Experience

Intuit

June 2026 – Present

Software Engineering Intern

Mountain View, CA

- Designed and shipped production backend features in **Java** for Intuit's enterprise data lineage platform, extending node resolution across additional asset types using **AWS Neptune** graph queries and structured error handling.
- Built **Model Context Protocol (MCP)** server tools exposing lineage **REST APIs** to **Claude AI** agents, enabling grounded asset resolution, lineage analysis, and ownership lookup through trusted production systems.
- Improved AI-tool reliability by defining clear behavior for successful, missing, ambiguous, and overly broad requests, preventing agents from guessing identifiers or returning misleading results.
- Optimized lineage-score processing with server-side **Gremlin** aggregation, preserving exact offending-node counts while limiting diagnostic payloads and application-side materialization.

Caltech Quantitative Finance Research Group

July 2025 – Present

Quantitative Trading Researcher

Pasadena, CA

- Built mean-reversion strategies in **Python/C** using **regime-switching models**, achieving **15% alpha** in backtests.
- Modeled **market microstructure** and order book dynamics to quantify cross-exchange arbitrage in simulation.
- Built a real-time **SQL**-based risk monitoring framework tracking **50+** portfolio metrics, drawdowns, and exposure.

Caltech First Year Success Research Institute (FSRI)

Aug 2024 – Sept 2025

Machine Learning Research Assistant

Pasadena, CA

- Engineered **U-Net** models in **Python/PyTorch/C++** for geological fracture detection, improving segmentation accuracy by **20%**.
- Integrated multispectral imagery and optimized preprocessing to improve model robustness across geological conditions.
- Developed automated labeling pipelines for **10,000+** UAV imagery samples using **Python** and Linux shell scripts.

CaltechDATA

Sept 2024 – June 2026

Research Data Software Engineer Assistant

Pasadena, CA

- Built **Python** automation pipelines to harvest and process **3,000+** research datasets through **REST APIs**.
- Implemented metadata analysis workflows to identify and reconcile missing datasets linked to Caltech publications.
- Supported data infrastructure used by **650+** faculty and researchers, improving research data reliability and preservation.

Medaase LifeCare Healthcare Platform

Jan 2024 – Present

Founder & Full-Stack Developer

Remote

- Built a full-stack healthcare platform that reduced appointment no-shows by **25%** through scheduling automation and reminders.
- Implemented real-time patient tracking and secure data handling for **100+ concurrent users**.
- Developed **Python**-based analytics dashboards to measure engagement and identify patient retention patterns.

Projects

Timely Tab | *JavaScript, HTML, CSS, Firebase, Mistral API*

Project Link

- Built and published a Chrome Web Store productivity app used across **250+** user sessions.
- Implemented **Firebase Auth/Firestore** real-time synchronization for reliable cross-device data persistence.
- Integrated the **Mistral API** to enable AI-driven content generation, improving app interactivity and personalization.

TraceQuery | *Python, FastAPI, React, Supabase, FAISS*

Project Link

- Built a full-stack **RAG** research assistant over uploaded PDFs with **FAISS** vector search and citation-backed responses.
- Designed semantic search, chat history, and citation tracing workflows with **Supabase/PostgreSQL** for reliable multi-user use.